

# ■■ DEFCON-D Security Experiment Report

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| Attribute                | Value      |
|--------------------------|------------|
| Experiment ID            | 18         |
| Experiment Name          | 232        |
| Traffic Profile          | NORMAL     |
| Expected Duration        | 30 seconds |
| Virtual Clients count    | 10         |
| Request Throttling delay | 0.5s       |

## ■ Host Performance Metrics Summary

|                            |                |
|----------------------------|----------------|
| Peak CPU Usage             | 49.3%          |
| Peak Memory Usage          | 74.9%          |
| Max Traffic Throughput     | 1 requests/sec |
| Max Concurrent Sockets     | 0 connections  |
| Average Processing Latency | 0.19 ms        |

## ■ Security Alerts Logged

| Severity | Alert Details                                                     | Attacker IP |
|----------|-------------------------------------------------------------------|-------------|
| RED      | Low IP distribution entropy: 0.00 bits (Targeted flood indicator) | 127.0.0.1   |
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| RED      | Low IP distribution entropy: 0.00 bits (Targeted flood indicator) | 127.0.0.1   |
| ORANGE   | Sudden traffic burst spike: 4 pps (Moving Average: 1.2 pps)       | 127.0.0.1   |
| RED      | Low IP distribution entropy: 0.00 bits (Targeted flood indicator) | 127.0.0.1   |
| ORANGE   | Sudden traffic burst spike: 4 pps (Moving Average: 1.6 pps)       | 127.0.0.1   |
| RED      | Low IP distribution entropy: 0.00 bits (Targeted flood indicator) | 127.0.0.1   |
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## ■ ■ Mitigation Actions Deployed

No mitigation overrides triggered during simulation window.

## ■ Experiment Conclusion & Defensive Action Summary

Based on collected metrics, the platform has successfully simulated client connections. Under elevated loading (Medium/Heavy profiles), request frequency and latency spikes trigger the anomaly engines. In Automatic defense mode, the mitigation systems successfully logged rate-limit offenses and temporary blacklist overrides to SQLite database, blocking the targeted flows. This experiment confirms the effectiveness of multi-tiered rate limiting and signature alerts in DDoS mitigation.